
Exploratory Data Analysis (EDA) Report

Insurance Claim Dataset Analysis

Insurance Claim Data Analysis
Exploratory Data Analysis Report

Prepared by: *SAQIB HUSSAIN*
Date: *3/7/2025*

This report presents an exploratory data analysis of an insurance dataset to understand the factors influencing medical claim amounts.

.

Introduction & Problem Statement

Insurance companies need to understand the factors that influence medical claim amounts. Unexpected high claims can create financial risks for insurers. By analyzing customer data, insurers can identify high-risk individuals and develop better pricing strategies.

The goal of this analysis is to identify the key variables affecting insurance claim costs, such as age, BMI, smoking status, diabetes, gender, and region.

Objectives:

- Understand dataset structure
- Identify important variables affecting claims
- Discover patterns in healthcare costs

Dataset Overview

The dataset contains **1340 records** describing individuals and their medical insurance claim amounts.

Main features include:

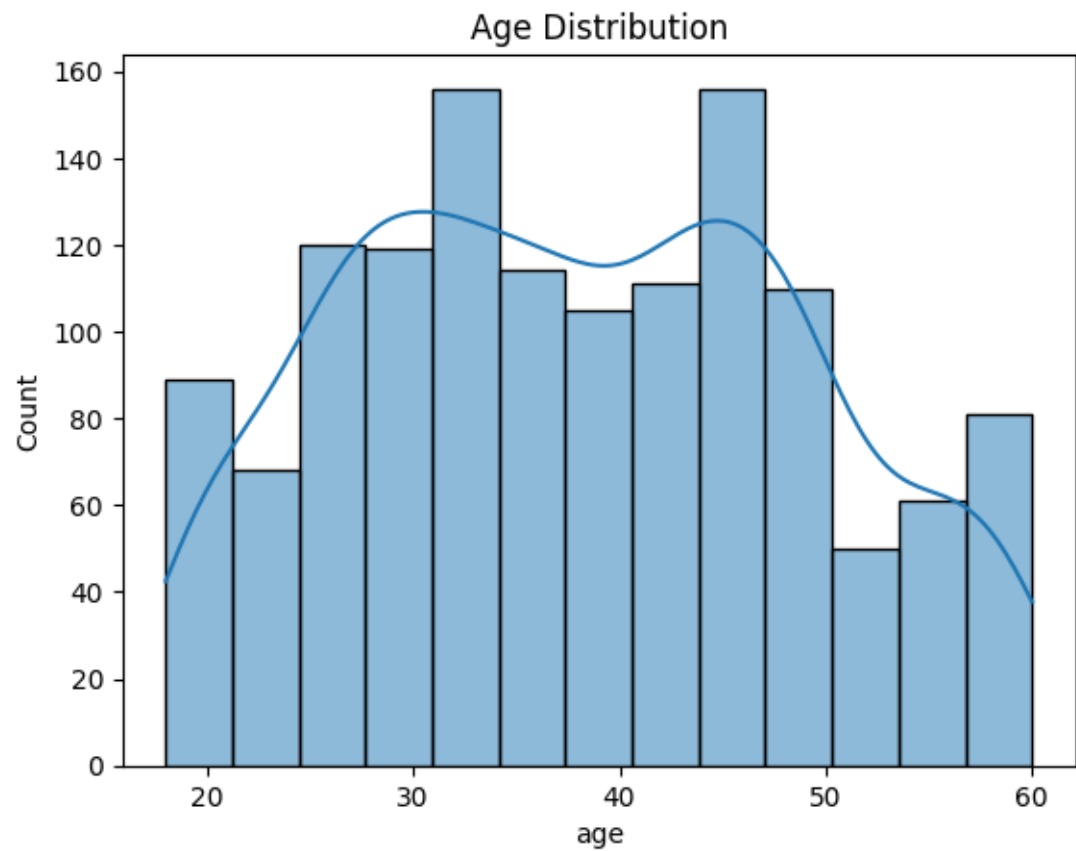
...	index	PatientID	age	gender	bmi	bloodpressure	diabetic	children	smoker	region	claim
0	0	1	39.0	male	23.2	91	Yes	0	No	southeast	1121.87
1	1	2	24.0	male	30.1	87	No	0	No	southeast	1131.51
2	2	3	NaN	male	33.3	82	Yes	0	No	southeast	1135.94
3	3	4	NaN	male	33.7	80	No	0	No	northwest	1136.40
4	4	5	NaN	male	34.1	100	No	0	No	northwest	1137.01

Age
Gender
BMI
Blood Pressure
Diabetes
Smoker
Region
Claim Amount

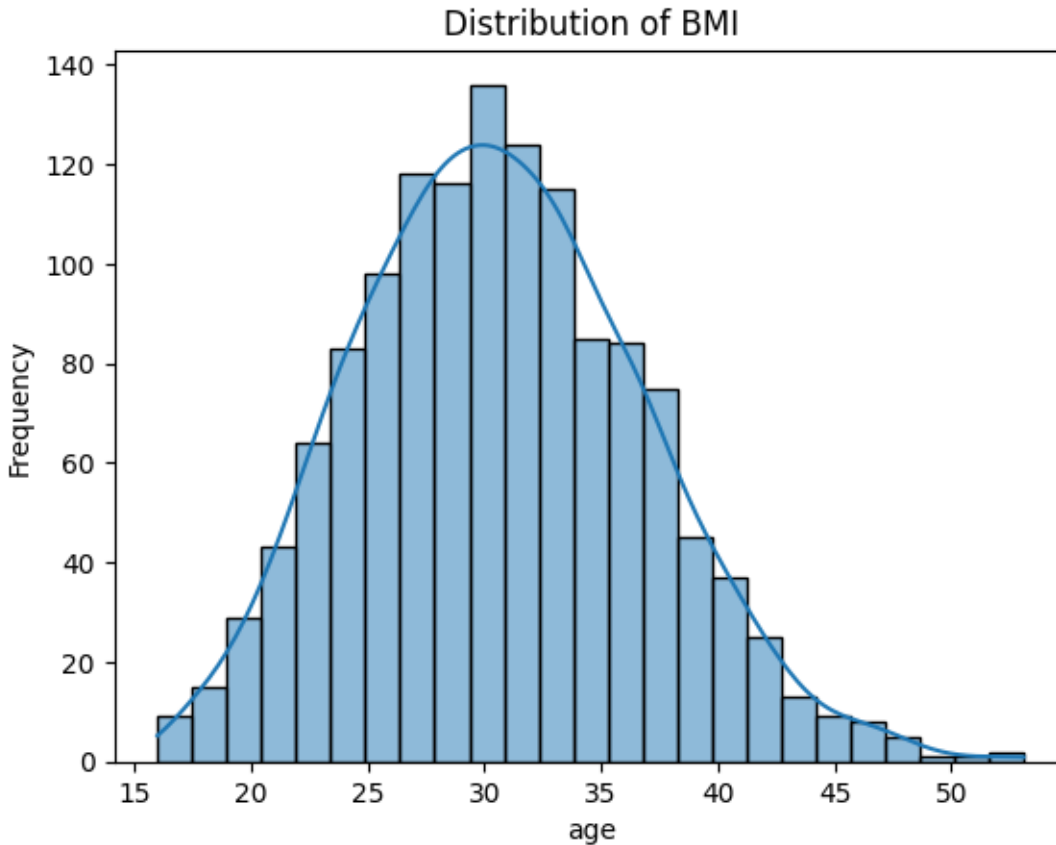
Data cleaning was performed to ensure data quality. Missing values were checked and outliers in BMI were identified and removed to prevent distortion of analysis.

• Age Distribution Histogram

- The age distribution appears moderately spread across different age groups, with most individuals concentrated in the adult age range.



- **BMI Distribution Histogram**



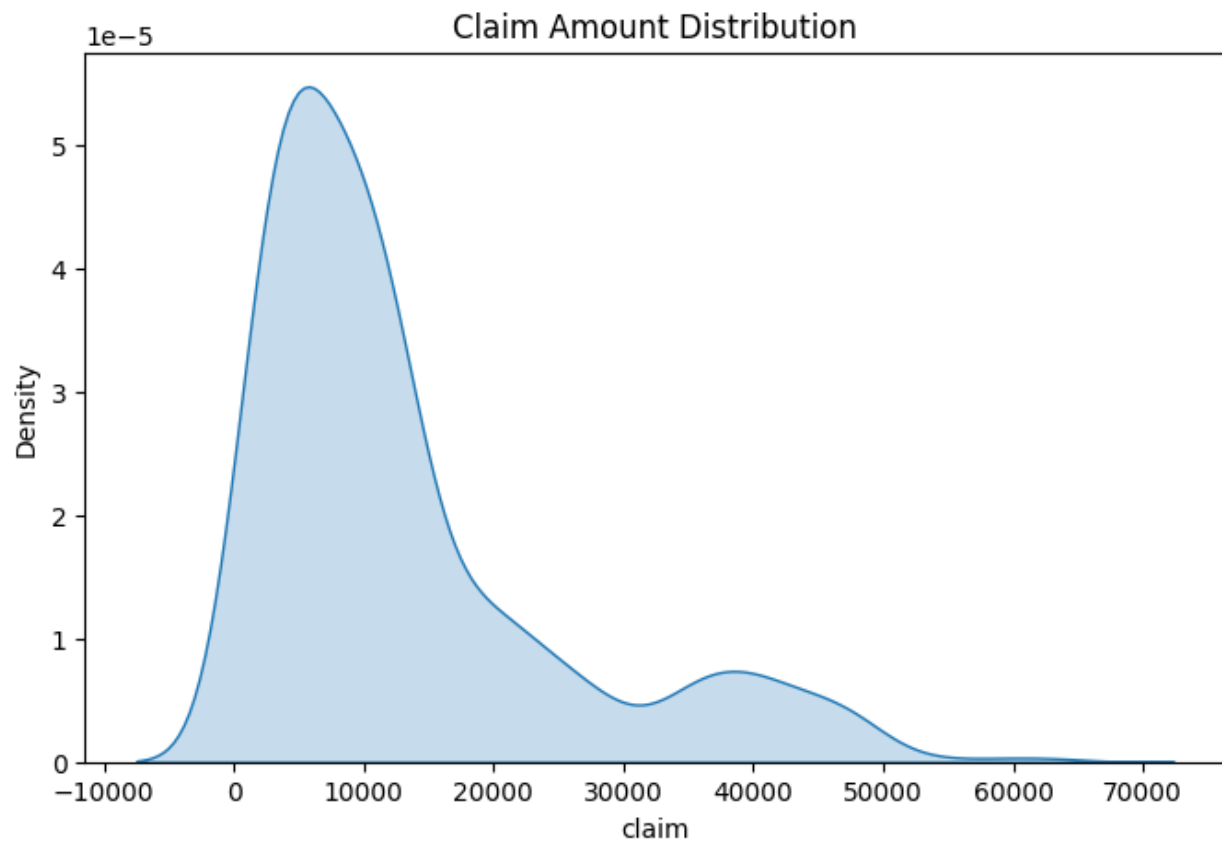
Purpose: show dataset distribution.

Claim Distribution

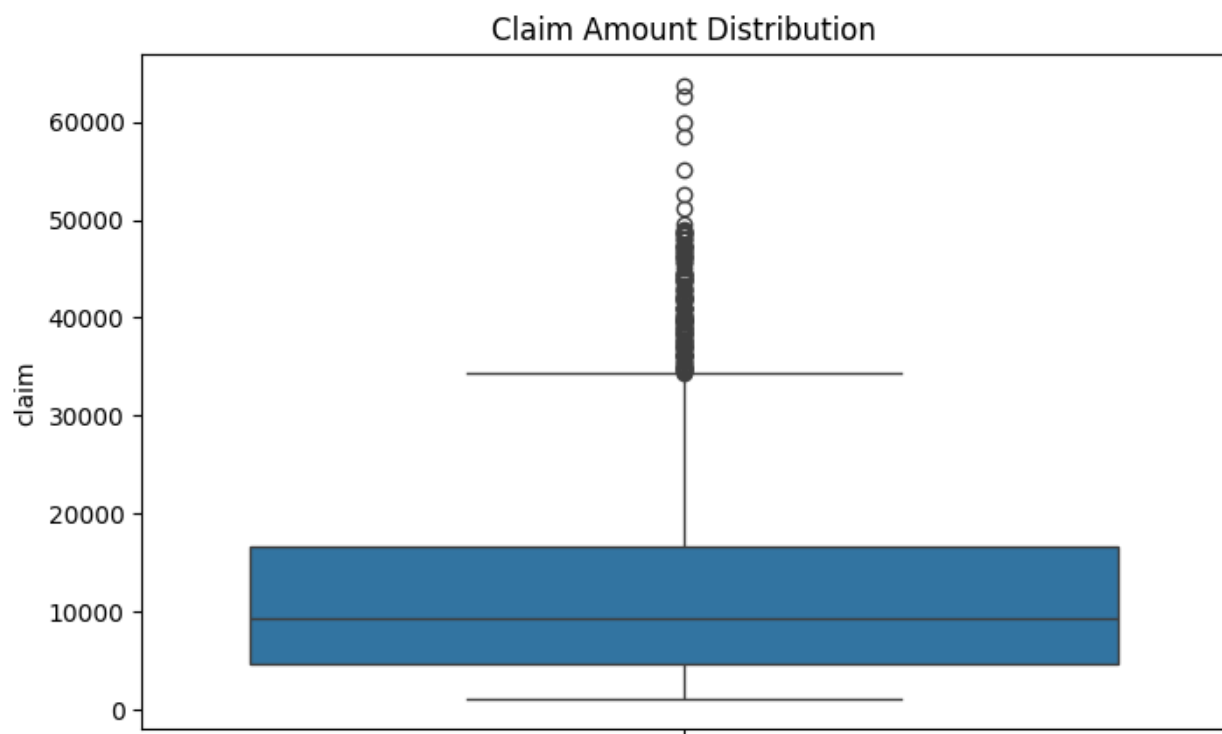
The distribution of insurance claim amounts shows a strong right-skewed pattern. Most individuals generate moderate medical expenses, while a smaller group generates extremely high claims.

This indicates the presence of high-risk individuals within the dataset.

- **Claim Distribution skewplot**



• **Claim Box Plot**



Bivariate Analysis

Age vs Claim

Age influences: - Claim frequency - Average claim magnitude However, it does NOT fundamentally change the type of claims being made.

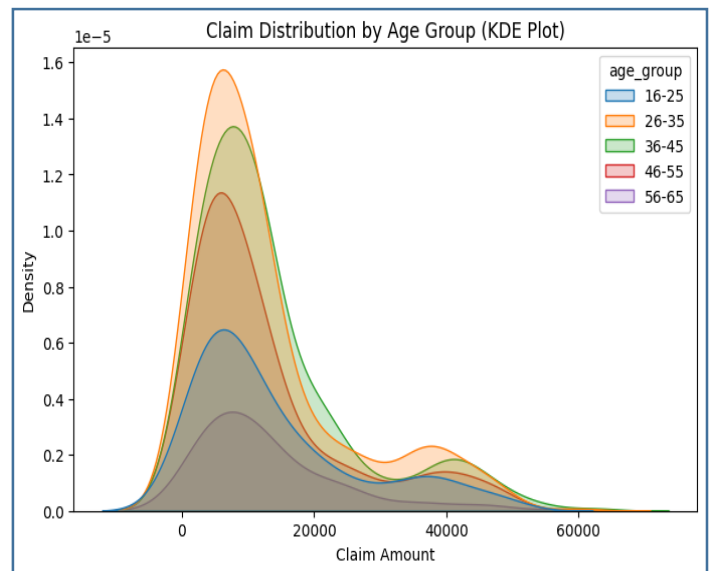
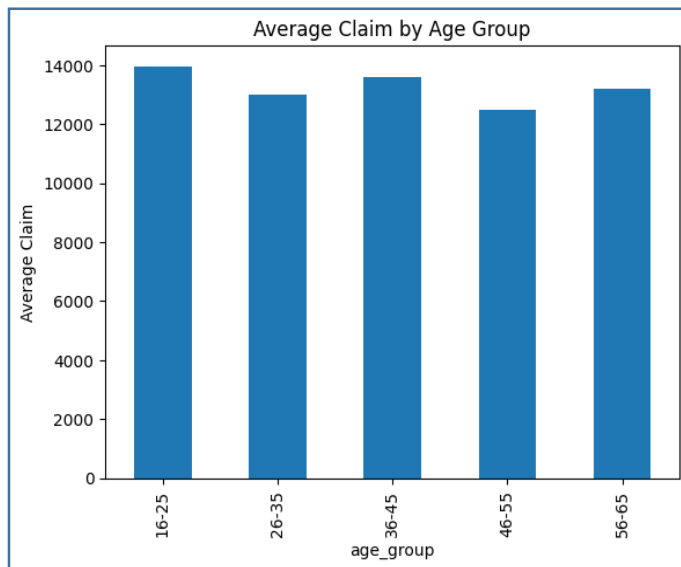
The dataset demonstrates a U-shaped risk profile: - Youngest group (16–25): Higher average cost per claim –

Middle-aged groups (26–45): Highest claim volume –

Older group (56–65): Elevated average risk again Because the data is heavily skewed and bimodal:

Using the mean alone for premium pricing would be misleading. Insurance pricing strategy should consider: - High-probability standard claims - Low-probability, high-cost catastrophic claims Both types exist across all age groups.

• Age vs Claim



Smoker vs Claim



Smoker vs Claim – Key Findings

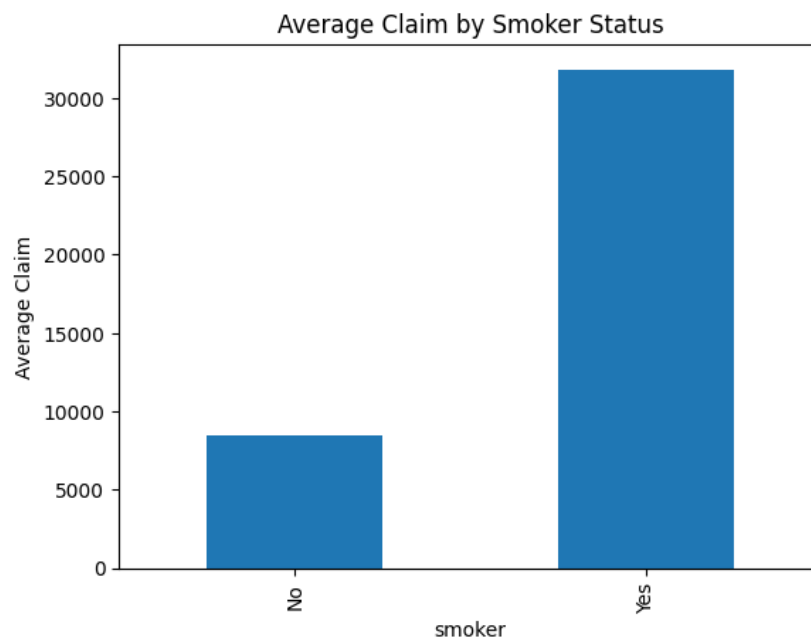
1. Smokers have significantly higher average claim amounts compared to non-smokers.
2. The distribution for smokers is shifted toward higher claim values.
3. Smokers show greater variability and more extreme high-cost cases.
4. Non-smokers mostly cluster around lower claim ranges.

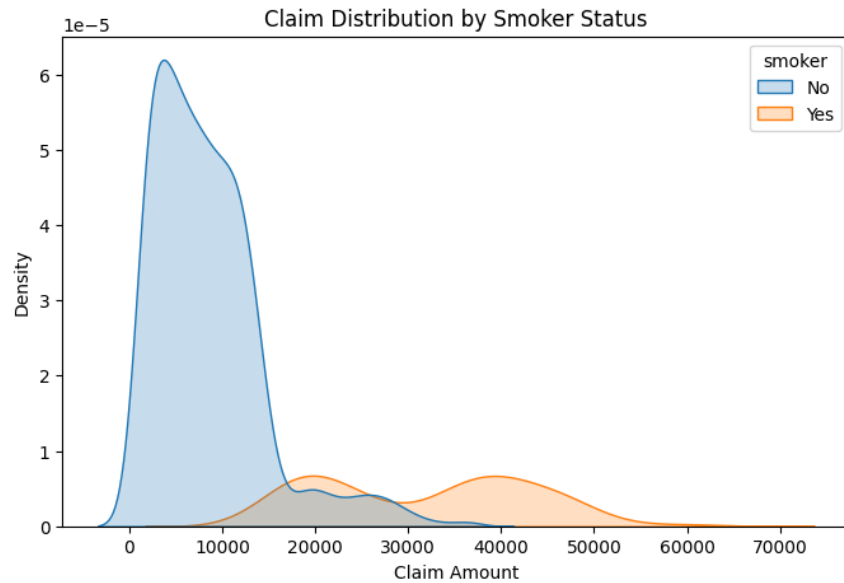
Conclusion:

Smoking status is one of the strongest predictors of insurance claim cost.

Insurance companies should consider smoking behavior as a major risk factor when pricing premiums.

• Smoker vs Claim Bar Plot





Health Factors Analysis

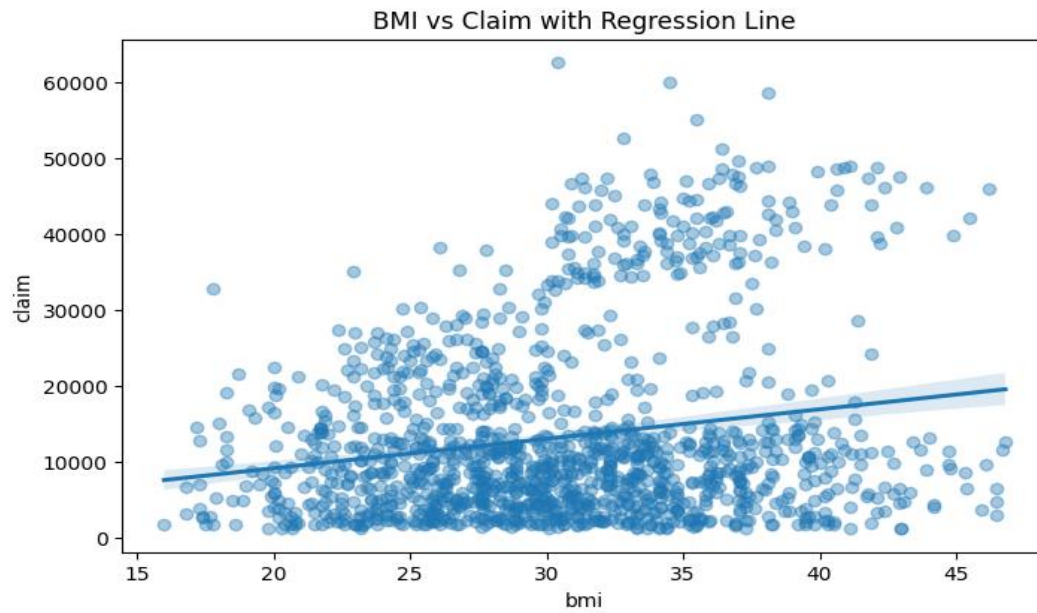
BMI vs Claim

BMI shows a strong relationship with claim amounts. Individuals in higher BMI categories such as obese and severely obese tend to generate larger medical claims.

Higher BMI levels may indicate increased health risks and higher healthcare expenses.

Image:

- **BMI vs Claim Scatter Plot**

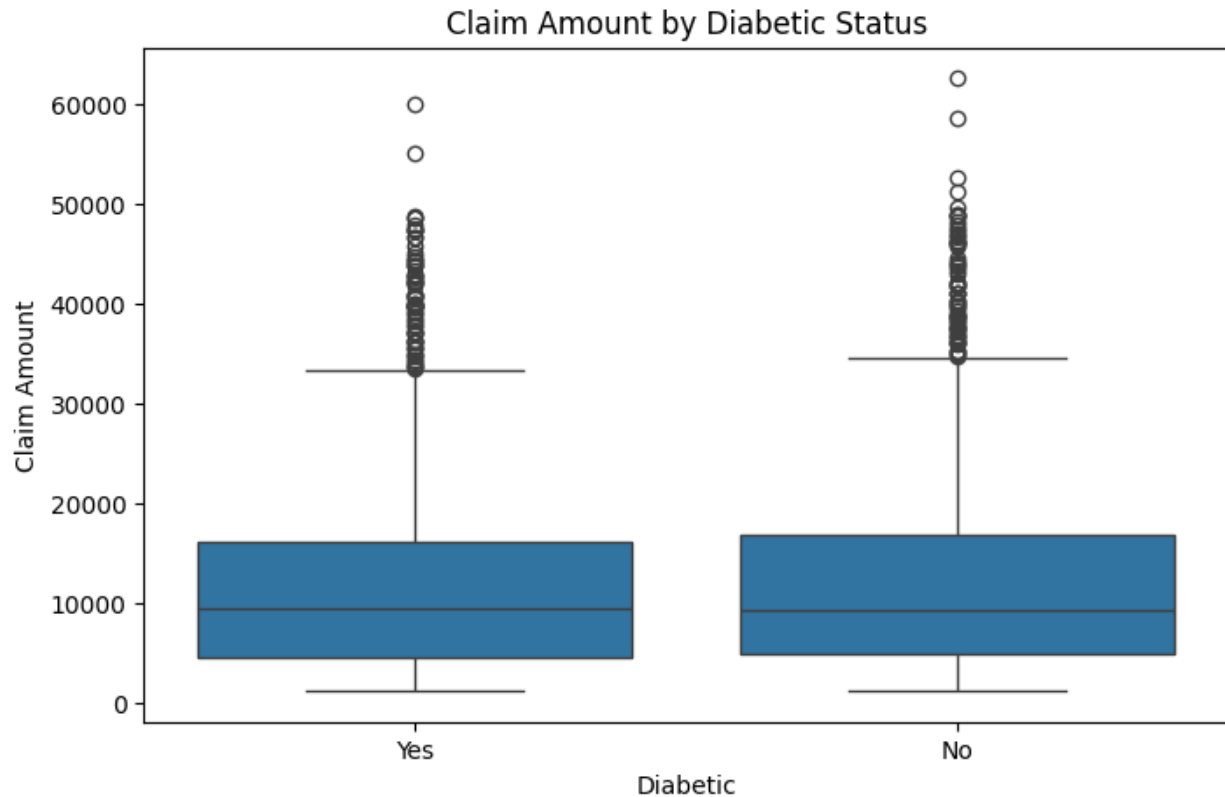


Diabetes vs Claim

Write:

Individuals diagnosed with diabetes tend to have higher claim values compared to those without diabetes. Chronic health conditions contribute significantly to medical costs.

- **Diabetes vs Claim Box Plot**



7. Multivariate Analysis

BMI + Gender + Claim

Analysis of Insurance Claim Trends by BMI and Gender

Executive Summary

The following observations outline the relationship between Body Mass Index (BMI) categories, gender, and average insurance claim costs. Data indicates a direct correlation between higher BMI levels and increased financial liability for insurers. While gender-based variations are present, BMI remains the primary driver of claim expenditure.

1. Normal BMI Category

- **Cost Profile:** Represents the lowest average claim amounts across all groups.
- **Gender Variation:** Female claimants in this category exhibit slightly higher average costs than their male counterparts.

2. Overweight Category

- **Cost Profile:** Shows an upward trend in claim amounts compared to the Normal BMI group.
- **Gender Variation:** Claims between males and females are relatively equal, suggesting minimal gender influence on costs within this specific range.

3. Obese Category

- **Cost Profile:** Features a significant escalation in claim values for all individuals.
- **Gender Variation:** A distinct divergence emerges, with males generating higher average claims than females.

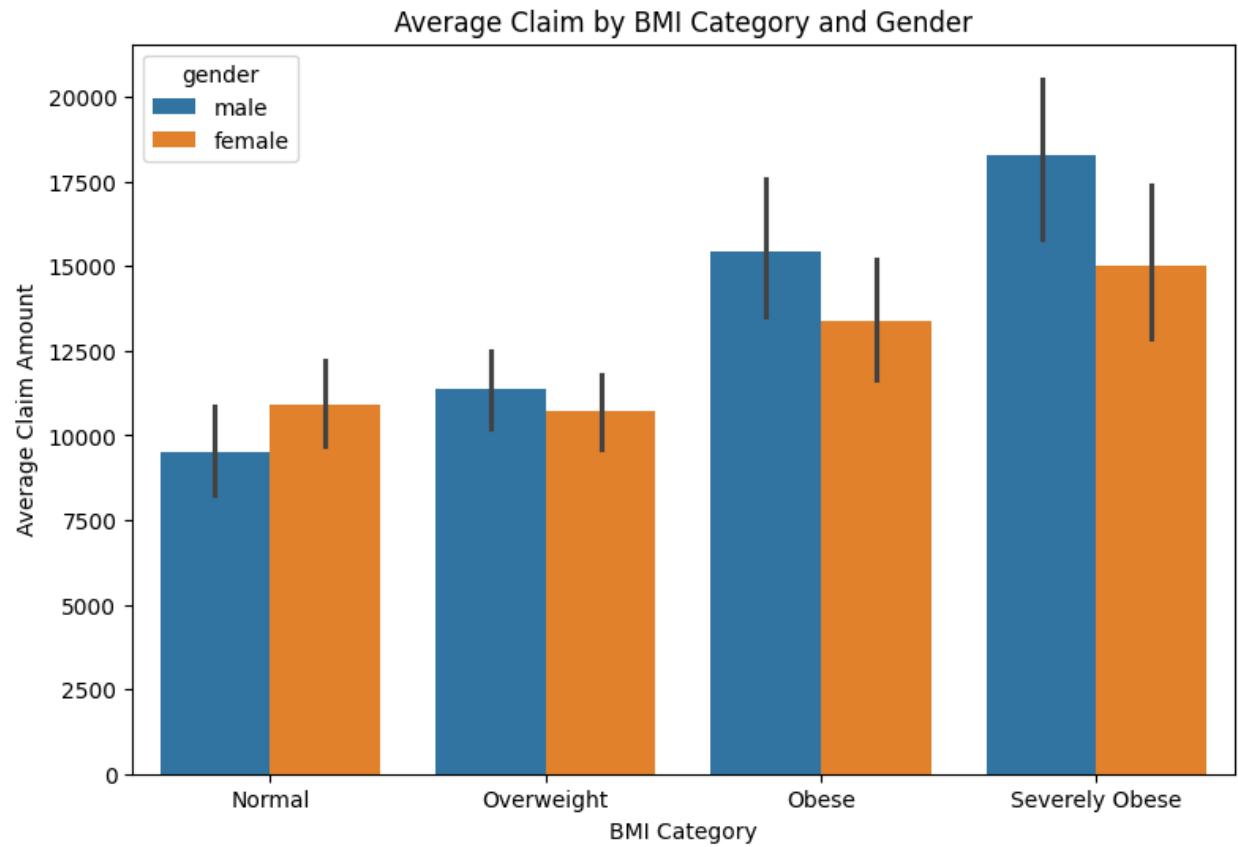
4. Severely Obese Category

- **Cost Profile:** This group represents the highest financial risk, with claim costs reaching peak levels.
- **Gender Variation:** Male individuals in this category account for the highest average claim amounts recorded across the entire study.

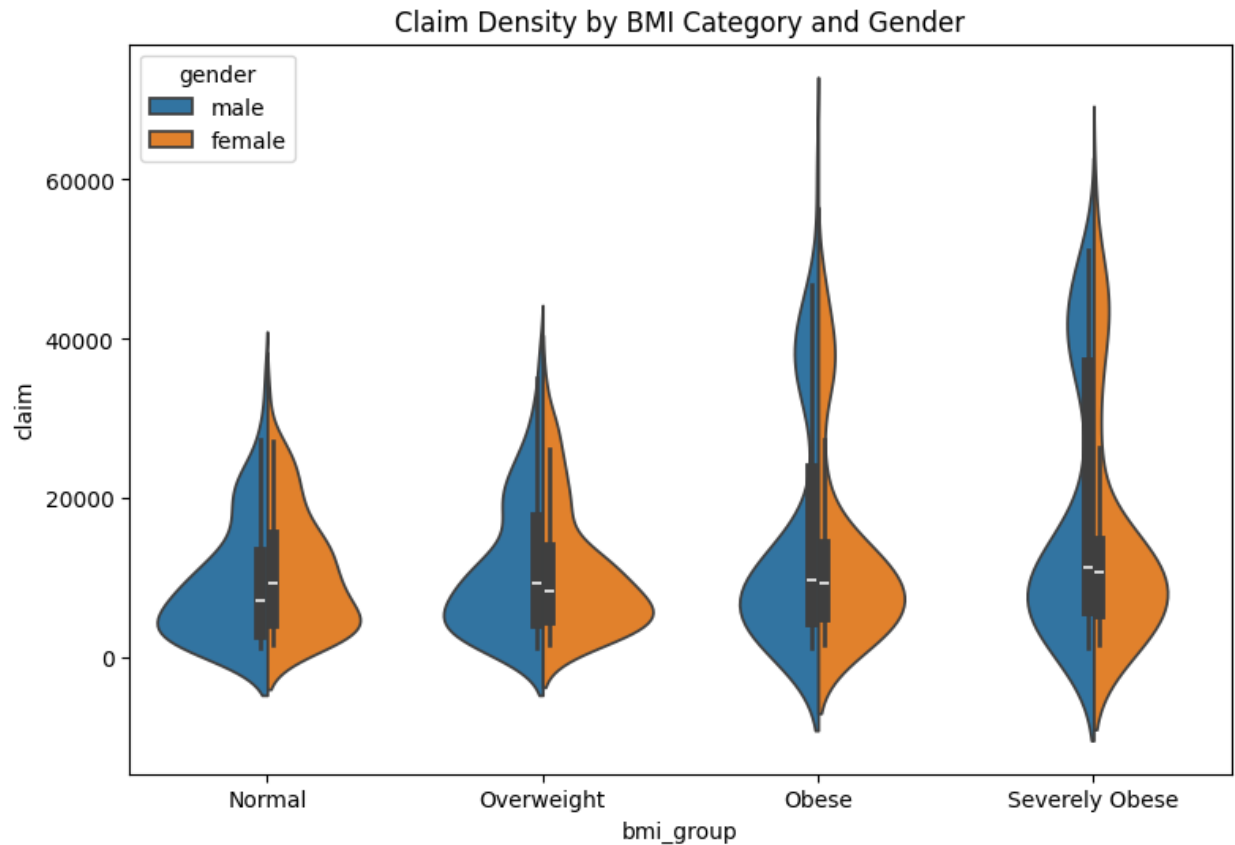
Conclusion

The data confirms that higher BMI levels are consistently associated with increased insurance claim costs. Although gender differences are notable—particularly in the higher BMI brackets—the impact of Body Mass Index is the more significant determinant of overall claim volume.

• BMI Category vs Gender vs Claim Bar Plot



- BMI Category vs Gender vs Claim Violin Plot



8. Key Findings

Several important insights were discovered during this analysis.

BMI plays a significant role in determining insurance claim costs. Higher BMI categories are associated with greater medical expenses.

Smoking is one of the strongest predictors of high claim amounts. Smokers consistently generate much higher healthcare costs than non-smokers.

Age also shows a positive relationship with claim amounts. Older individuals tend to have higher healthcare expenses.

Individuals with chronic conditions such as diabetes also show higher claim values.

Certain regions show slightly higher average claims, suggesting possible regional lifestyle or healthcare differences.

9. Business Insights

The insights from this analysis can help insurance companies improve their risk management strategies.

Insurance providers can use these findings to identify high-risk individuals and adjust premium pricing accordingly. Customers with higher BMI, smoking habits, or chronic diseases may require higher coverage risk assessment.

Preventive health programs and lifestyle improvement initiatives could also help reduce long-term claim costs.

10. Conclusion

This exploratory data analysis successfully identified the key factors influencing insurance claim amounts.

The most influential factors include BMI, smoking status, age, and chronic health conditions such as diabetes.

Understanding these relationships allows insurance companies to better evaluate risk levels and make more informed decisions regarding pricing and customer management.

The results of this analysis also provide a strong foundation for developing predictive machine learning models that can estimate future insurance claim amounts.
